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RoomieSplit

A Full-Stack Web Application for Shared Living Finance and
Chore Management.

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The Problem

Roommates often share expenses such as rent, groceries, utilities, and subscriptions, while also dividing household chores.

Managing these responsibilities manually creates several issues:

Difficulty tracking
who paid what

Confusion over who
owes whom

Poor chore
coordination

Tension between
housemates

RoomieSplit was created to solve this problem in one centralized system.

Our Solution

RoomieSplit is a full-stack single-page web application designed to help roommates:



Centralized
Management



Automated
Balances



Tracking &
Scheduling



Role-Based
Access



Assigned
Chores



Admin
Overview

The goal is to reduce mental bookkeeping and make shared living more organized.

Main Features

Multi-group
support

Shared Expense
Splitting

Chore
Management

Settlement
Tracking

Calendar-based
Tracking

Group Admin
Controls

System Admin
Dashboard

Email Invitation
Flow

Dark &
Light Themes

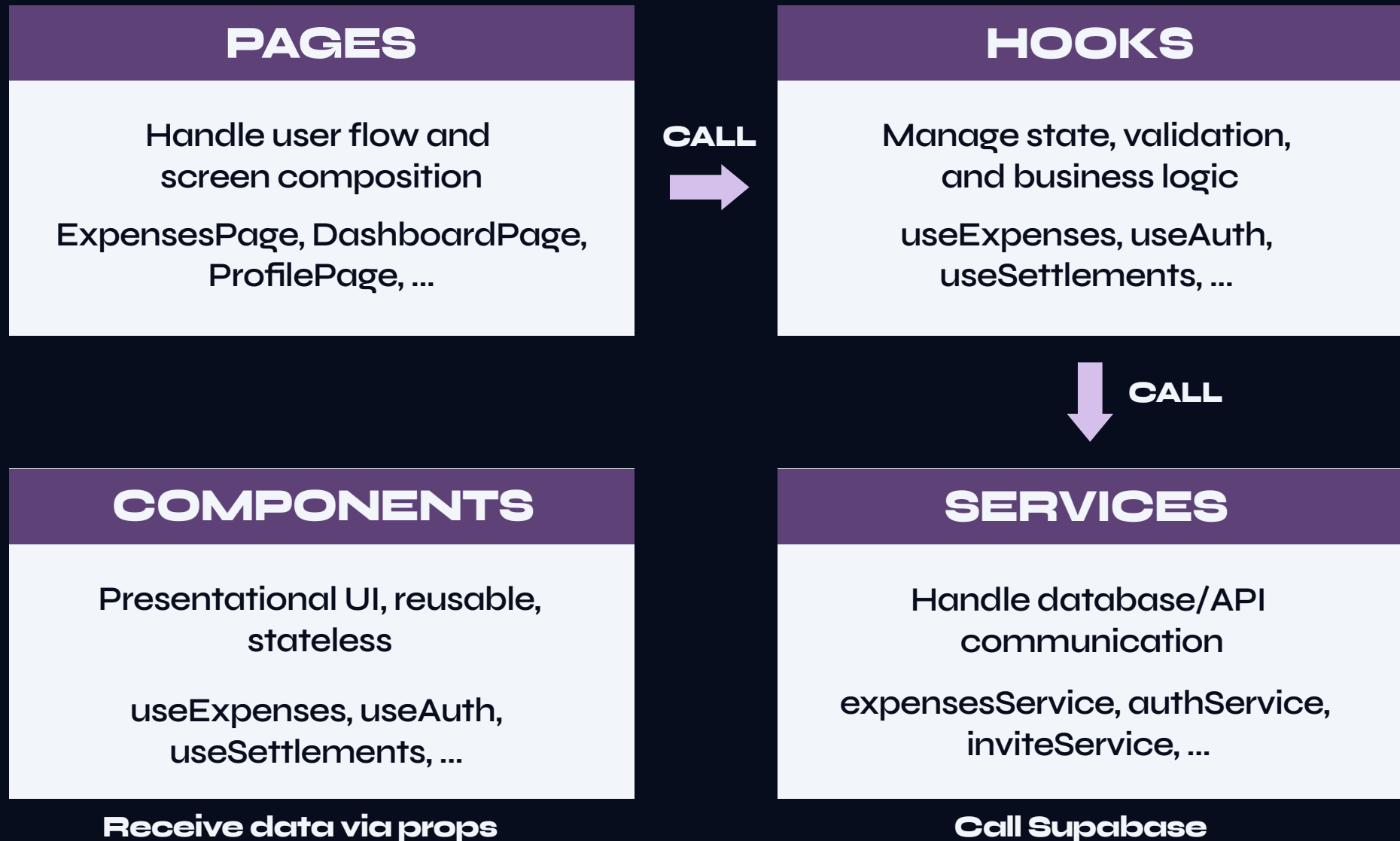
This makes the app useful for both everyday users and administrators.

Tech Stack

FRONTEND	BACKEND
React 19	Supabase
TypeScript	PostgreSQL
React Router DOM	Supabase Auth
Tailwind CSS	Supabase Storage
Vite	RPC Functions

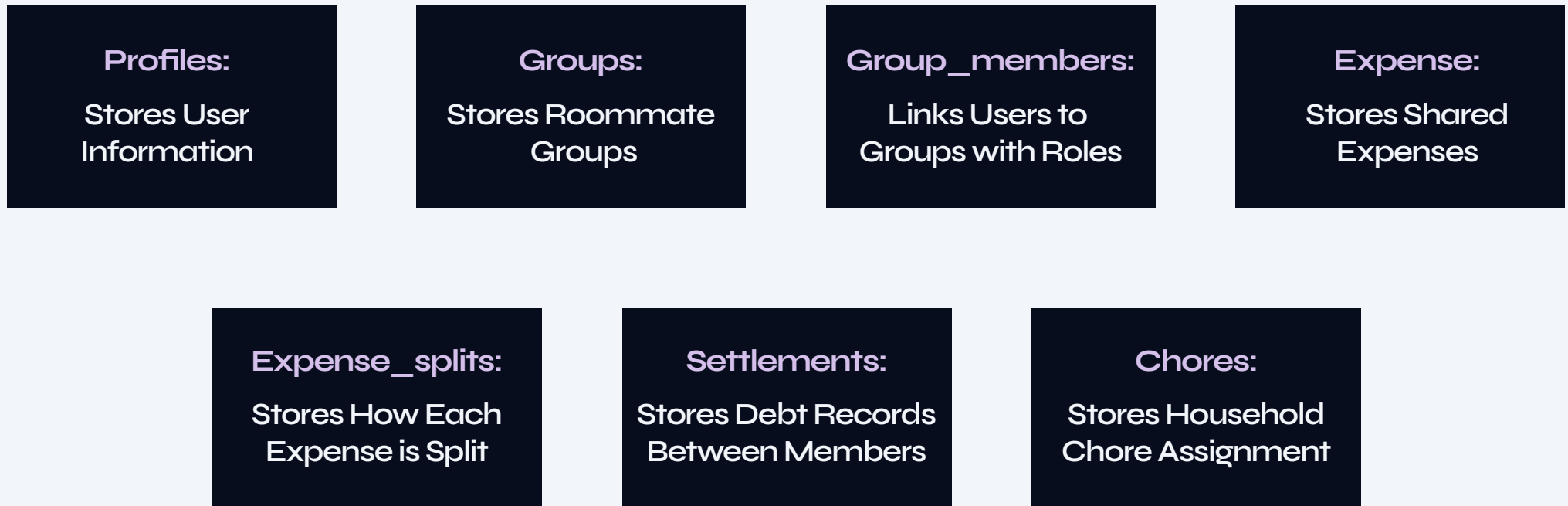
TESTING
Supabase
PostgreSQL
Supabase Auth

System Architecture



Database Structure

The system uses 7 main database tables:



These tables work together to support the full roommate workflow from joining groups to settling payments.

User Roles & Security

REGULAR MEMBER	GROUP ADMIN	SYSTEM ADMIN
View expenses, balances, chores	All member permissions	View all users, groups, and records
Create expenses	Add or remove members	Deactivate users
Record their payments	Change member roles	Archive groups
Complete chores	Manage the group	Access the admin dashboard

Security is enforced using Supabase Row Level Security policies, so access is controlled directly at the database level.

Expense Workflow

This entire workflow is handled atomically using an RPC function, which ensures data consistency.

1

The user enters
description, amount,
payer, date, and
split members

2

The system validates
the data

3

The expense is
stored in the database

4

Split records are
created

5

Settlement records
are automatically
generated for members
who owe money

Example: If one of 3 members pays \$30 for groceries, the app automatically generates a payment of \$10 for each of the other members

Implementation Highlights

Currency Calculations

The app uses cent-based integer math to avoid floating-point errors.

Automatic Settlement

Whenever an expense is created, the system creates the matching debt records automatically.

Chore Icon Assignment

Chores are given icons automatically using keyword based matching with deterministic fallback logic.

Theme Persistence

Dark/light mode is stored in localStorage and also respects the user's operating system preference.

Frontend Design & User Experience

INTERFACE

Clean and modern

Responsive sizes

Easy to navigate

Consistent across pages

KEY UI FEATS

Reusable components

Collapsible sidebar

Top navigation bar

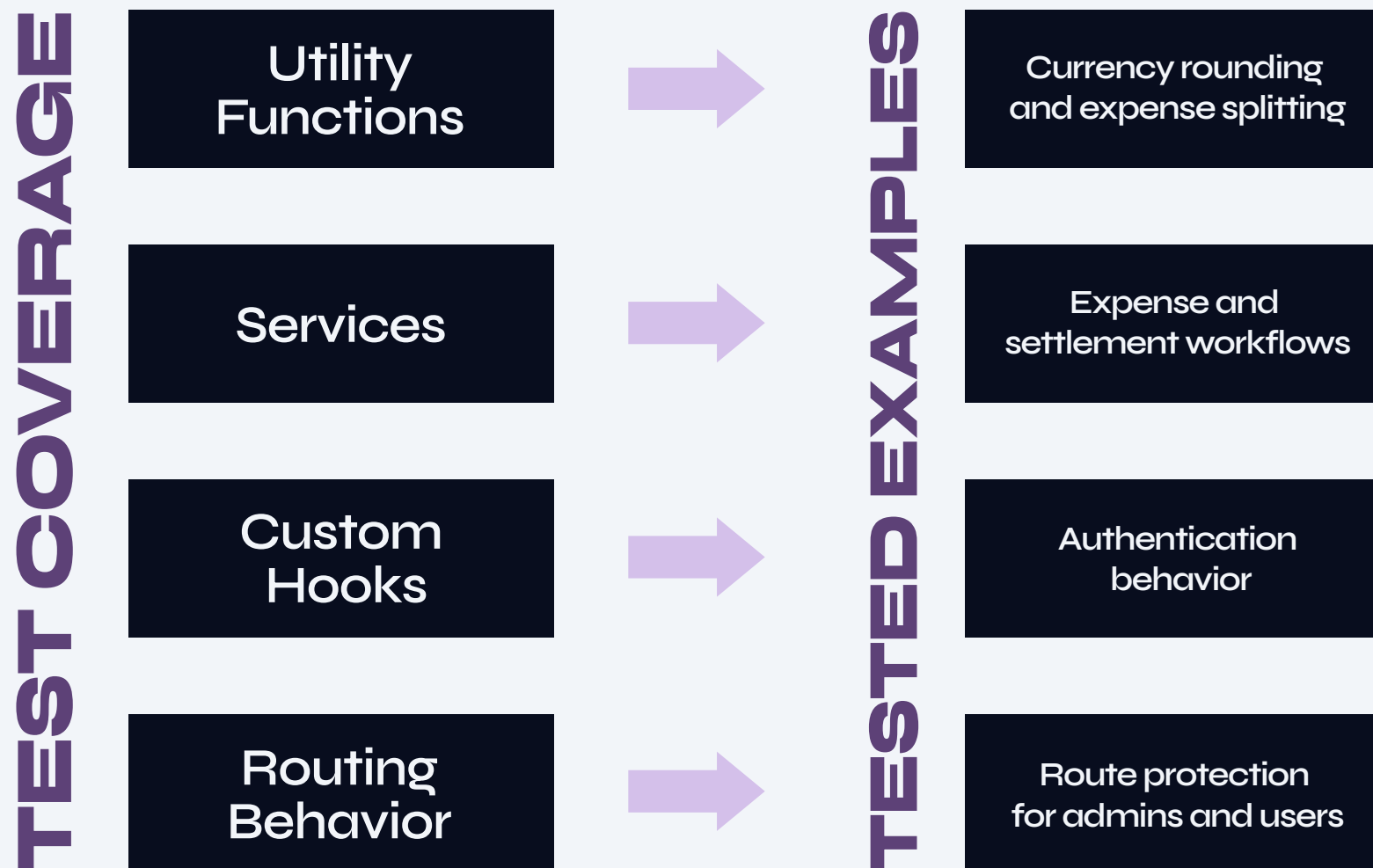
Theme toggle

Calendar tracking

The app focuses on usability while keeping the design professional and minimal.

Testing and Quality Assurance

The project includes a comprehensive test suite with 152 passing tests.



Limitations & Future Improvements

LIMITATIONS

No real-time updates between users

No recurring expense automation

No receipt support

No built-in notification system

No CSV/PDF export

No mobile app

IMPROVEMENTS

Add Supabase Realtime

Add recurring expenses rotation

Add email receipts

Add notifications

Add export features

Develop a mobile version

These improvements would make the system more scalable and production-ready.

Conclusion

RoomieSplit successfully delivers a complete full-stack solution for managing shared living by combining:

Expense splitting
and settlement tracking

Chore management
& calendar visualization

Role-based access
control and secure
data handling

Scalable architecture
using React and
Supabase

The project demonstrates strong technical design, clean architecture, and real-world applicability. It provides a solid foundation that can be extended into a production-ready system.



THANK YOU!